Manuel **Di Agostino**

PhD student in Mathematics

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Education

University of Parma
Parma, Italy

PhD in Mathematics (expected) — *Nov. 2026 - Present*

University of Parma
Parma, Italy

Master's Degree in Computer Science — *Sep. 2024 - Jul. 2026*

Grade: 110/110 *cum laude*
Title: "Towards Statically Reasoning About R Vectors"
Supervisor: Prof. Vincenzo Arceri
Co-supervisor: Florian Sihler (PhD Candidate)
Keywords: *Static Analysis, Abstract Interpretation, R Language, Software Verification*

University of Parma
Parma, Italy

Bachelor's Degree in Computer Science — *Sep. 2021 - Jul. 2024*

Grade: 110/110 *cum laude*
Title: "Valutazione sperimentale sull'individuazione automatica di errori di programmazione nel codice generato da LLM"
Supervisor: Prof. Enea Zaffanella
Co-supervisor: Prof. Vincenzo Arceri
Keywords: *Static Analysis, AI, LLM, Software Verification and Validation*

Interests

My main research interest lies in the mathematical foundations of static program analysis, particularly the theory and application of abstract interpretation. I enjoy designing and implementing abstract domains that capture complex program behaviors — recently, this has involved developing a domain for R vectors that handles indexing, recycling, and missing values. Beyond this core focus, I am broadly drawn to theoretical computer science, formal methods, and mathematical logic. I also enjoy reasoning about algorithms and working on language processing tools such as parsers.

Teaching

University of Parma
Parma, Italy

Tutor ("Adotta una matricola" program) — *Sep. 2025 - June 2026*

- Mentorship program for first-year students: providing guidance and support to help new students adapt to university life and academic requirements.
- Program details available at `https://www.unipr.it/bandi/bando-200-assegni-di-tutorato-aa-20252026`.

University of Parma
Parma, Italy

Tutor (Logic and Discrete Structures) — *Sep. 2025 - Jan. 2026*

- Academic support: provide assistance to first-year undergraduate students in understanding the theoretical concepts of logic and discrete mathematics.
- Practical exercises: guide students in solving problems related to set theory, algebraic structures, natural and structural induction, as well as propositional and predicate logic, helping them consolidate the necessary practical skills.
- Providing resources: offer supplementary materials and useful resources for further exploration of the topics covered in the course (available at `https://github.com/unipr-org/tutorati/tree/main/EdL`).

University of Parma
Parma, Italy

Tutor (Algorithms and Data Structures) — *Feb. 2025 - June 2025*

- Conceptual support: assist students in understanding fundamental algorithmic concepts, data structures, and their applications.
- Problem-solving sessions: guide students through exercises involving sorting, recursion, dynamic programming, and common data structures like stacks, queues, linked lists, trees, and graphs.
- Supplementary materials: provide additional learning materials and practical examples to reinforce comprehension (available at `https://github.com/unipr-org/tutorati/tree/main/ASD`).

University of Parma
Parma, Italy

Tutor (Logic and Discrete Structures) — *Sep. 2024 - Jan. 2025*

Publications

2026 **M. Di Agostino**, F. Sihler, V. Arceri, O. Gerstl, M. Tichy**,** Towards Statically Reasoning About R Vectors *NSAD 2026*

Talks

Towards Statically Reasoning About R Vectors *NSAD 2026*

Presenter *2026*

Academic Service

2027 **Artifact Evaluation Committee**, VMCAI 2027

Grants

Research participant, "LLMs Meet Static Analysis: improving quality and reliability of AI-generated code"

ISCRA Project (class C), Cineca *Jan. 2024 - June 2024*

Principal Investigator: Prof. Vincenzo Arceri. The goal of the project is to conduct an extensive quality and safety evaluation of the code generated with some of the most popular and open-source LLMs employing static analyzers, that can detect vulnerabilities and run-time errors statically, without executing the code. Once this information is available, it will be included in the code-generation task, to guide the LLM itself to produce a more precise and safe output, in which static analysis is somehow introduced in the pipeline of the code-generation task.

Schools and Workshops

2026 **Invited Participant**, CSV 2026, 5th Challenges of Software Verification Symposium *Venice, Italy*
2025 **Invited Participant**, CSV 2025, 4th Challenges of Software Verification Symposium *Venice, Italy*
2024 **Participant**, Lipari Summer School on Abstract Interpretation *Lipari, Italy*
2024 **Invited Participant**, CSV 2024, 3rd Challenges of Software Verification Symposium *Venice, Italy*

Additional experience

University of Ulm *Ulm, Germany*

Internship (Erasmus+ SMT) *March. 2026 - May. 2026*

- Co-authored a paper on abstract interpretation for R vectors, defining formal semantics for vector operations with support for positive/negative/logical indexing, recycling rules, and NA handling.
- Designed and implemented abstract operations for vector manipulation (recycle, select, update) with proper handling of must/may attributes, enabling precise static analysis of R code.
- Implemented the vector abstract domain in flowR (available at `https://github.com/manueldiagostino/flowr/tree/r-vectors/src/abstract-interpretation/vector`).

University of Parma *Parma, Italy*

Internship *Oct. 2023 - Jan. 2024*

- Examined the Mopsa static analyzer (`https://mopsa.lip6.fr/`) through a series of benchmarks (coreutils-benchmarks, juliet-benchmarks).
- Integrated the PPLite library (`https://github.com/ezaffanella/PPLite`) into the framework.
- Created a docker container to automate the installation (`manueldiagostino/mopsa:pplite`).

Tools and Software

Vector Abstract Domain for flowR *From October 2025*

A TypeScript implementation of an abstract domain for R vectors within the flowR static analyzer. This module performs compile-time inference of vector properties including length, value ranges, type, and attributes. It supports complex R operations such as concatenation, positional/logical selection and update, recycling rules, and NA-aware arithmetic. The implementation follows the formal semantics defined in an accompanying research paper on abstract interpretation for R, enabling precise static analysis of vector manipulation code in R programs (`https://github.com/manueldiagostino/flowr`).

Tree-sitter grammar for StrictDoc *From June 2025*

A Tree-sitter grammar for the StrictDoc language. This tool enables robust and efficient parsing of StrictDoc files, allowing for features like syntax highlighting, code navigation, and structural analysis in compatible text editors and IDEs. The grammar is designed to work with the Tree-sitter parsing framework, improving the developer experience when working with StrictDoc requirements files (`https://github.com/manueldiagostino/tree-sitter-strictdoc`).

Deep Neural Network Library *From May 2024*

A flexible C++ library for building, training, and using deep neural networks. Modular, easy-to-use, and high-performance. The library supports various neural network architectures, efficient training tools, and robust inference methods (`https://github.com/unipr-org/deep-neural-network`).